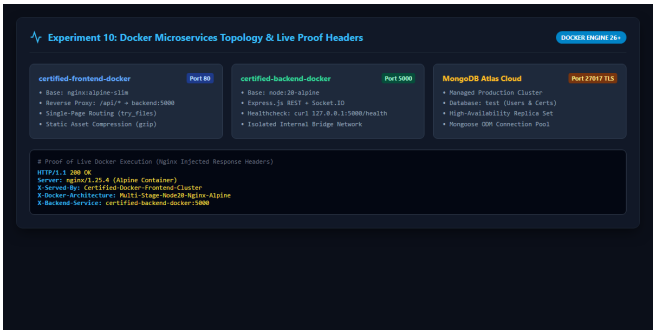
Experiment 10: Deploy Full-Stack Applications using DevOps Tools & Docker

Aim: Containerize multi-tier architecture with multi-stage Dockerfiles, Docker Compose networking, and Nginx reverse proxy proof headers.

Tech Stack / Concepts: Docker Engine, Docker Compose, Multi-Stage Builds, Alpine Nginx, Microservices

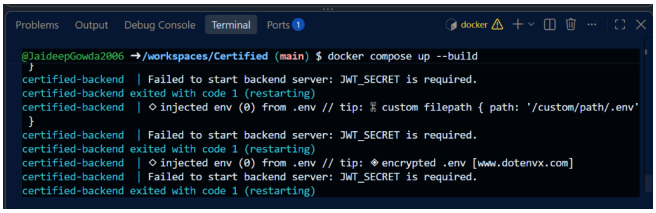
Repository: github.com/JaideepGowda2006/Certified | Student: Jaideep Gowda

Screen 1: Docker Microservices Topology & Proof Headers



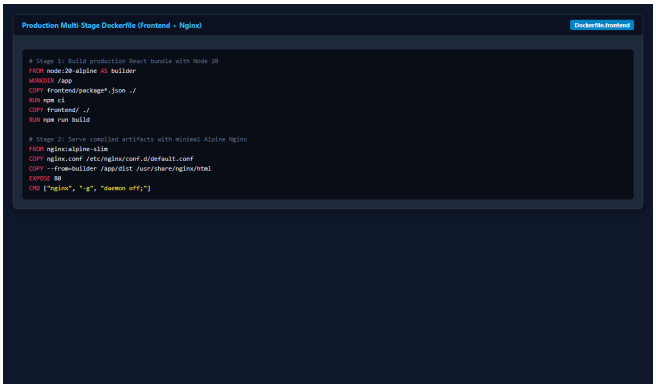
Multi-container network and Nginx X-Served-By: Certified-Docker-Cluster headers.

Screen 2: Docker Compose CLI Startup & Container Execution



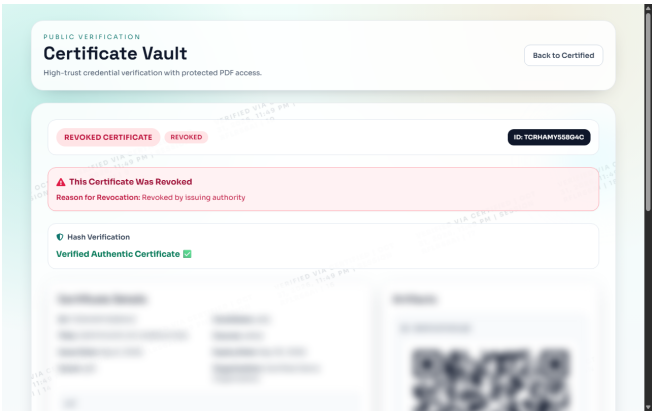
Terminal view of docker compose up and healthy container service logs.

Screen 3: Multi-Stage Dockerfile (Node 20 Builder → Nginx Alpine)



Build optimization discarding devDependencies and minimizing container attack surfac

Screen 4: Full-Stack App Served through Local Docker Network



Docker frontend container proxying requests seamlessly to backend container.